
Tactile Graphics Finger Pattern Analysis System

An End-to-End Research Framework for Analyzing How Blind & Visually Impaired (BVI) Participants Explore Tactile Graphics Through Touch

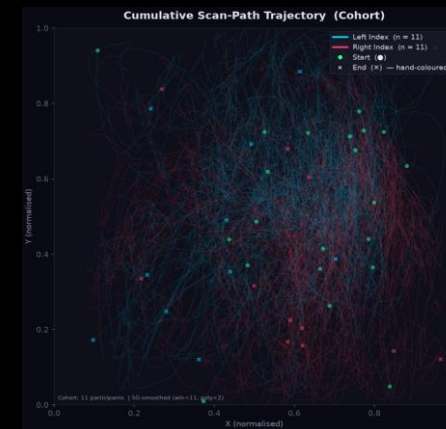
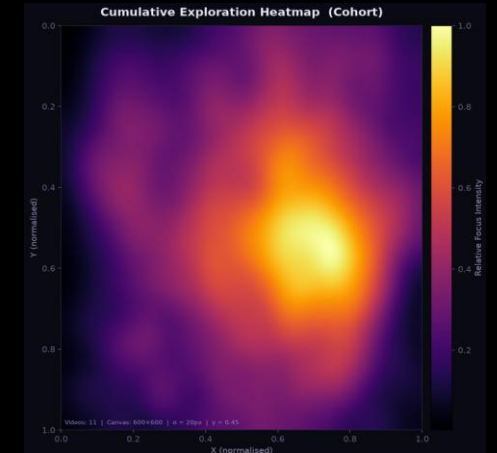
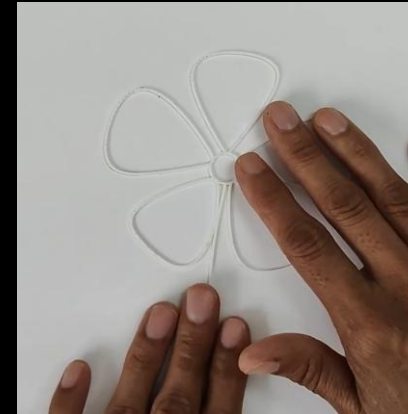
YOLO Fine-Tuning

Behavioral Analytics

Interactive Dashboard

Introduction

- Understanding how Blind and Visually Impaired (BVI) individuals explore tactile graphics is essential for improving accessible educational materials and assistive technologies.
- We developed the **Tactile Graphics Finger Pattern Analysis System (TGFPAS)**, an integrated application that automatically analyses finger exploration behaviour from participant videos using fine-tuned YOLO models and region-based behavioural analysis.
- The system generates detailed visualizations, including heatmaps, trajectories, transition graphs, and exploration paths, while separately analysing correct and incorrect exploration sessions as well as left-hand, right-hand, and combined-hand movements.
- This provides researchers with valuable insights into tactile perception and exploration strategies through a fully automated and scalable analysis pipeline.



Beyond Detection — Understanding How BVI Users Explore

Tactile graphics are the primary medium through which BVI individuals access diagrams, maps, and figures. Simply detecting where a finger is tells us very little — the real research questions are behavioral:

HOW

do participants explore a tactile diagram — the strategy, path, and rhythm of exploration?

WHICH

salient regions receive attention, and in what order are they first touched?

WHAT

distinguishes successful identification from unsuccessful attempts?

HOW DO

the left and right hands cooperate during bimanual exploration?

TGFPAS answers these with an automated, reproducible pipeline — from model training to publication-ready analytics.

One Desktop Application · Four Independent Modules



Built with **Python + PySide6** as a single research desktop app. Each tactile graphic gets its **own model (best.pt)** and **own region map (JSON)** — making the framework instantly scalable to new graphics without touching application code.

Data Collection & Cohort Design

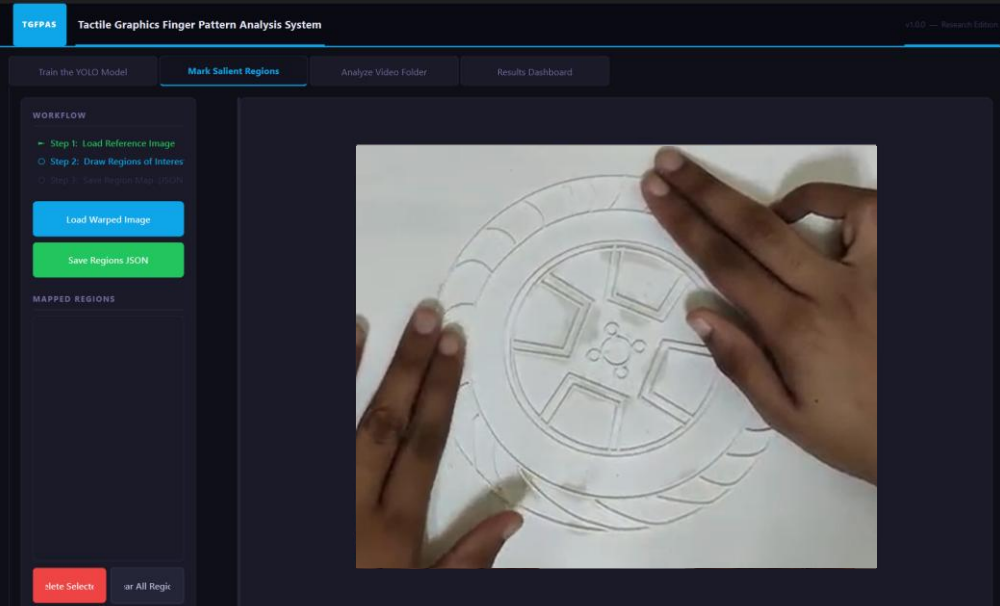
- Videos of BVI participants exploring tactile graphics were provided by the research mentor.
- Each participant attempted to identify the graphic purely through touch.
- Videos were split into two cohorts based on the outcome of identification — kept strictly separate through the entire pipeline.
- Every tactile graphic maintains its own independent dataset, its own fine-tuned model, and its own region map.

CORRECT**6**

videos — Tire graphic
(successful identification)

INCORRECT**20**

videos — Tire graphic
(unsuccessful identification)



A BVI participant's hands on the raised-line tactile graphic, loaded as the reference image inside TGFPAS.

Why two cohorts? Comparing them exposes the behavioral signature of successful exploration.

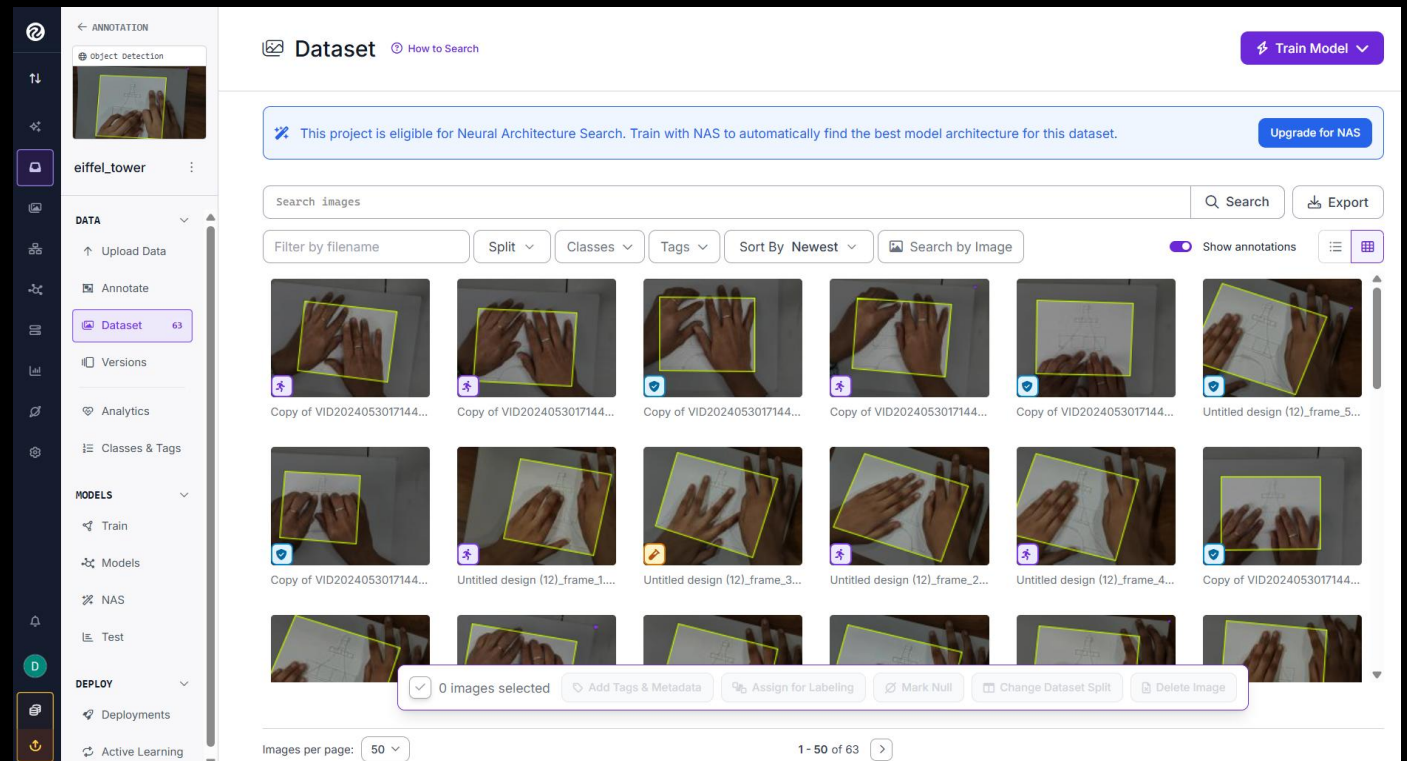
Preparation for Model Training

The first module of the application focuses on creating a customized object detection model for finger tracking. Since a generic pre-trained YOLO model is not sufficiently accurate for detecting finger positions during tactile exploration, a dedicated fine-tuning process was developed for every tactile graphic.

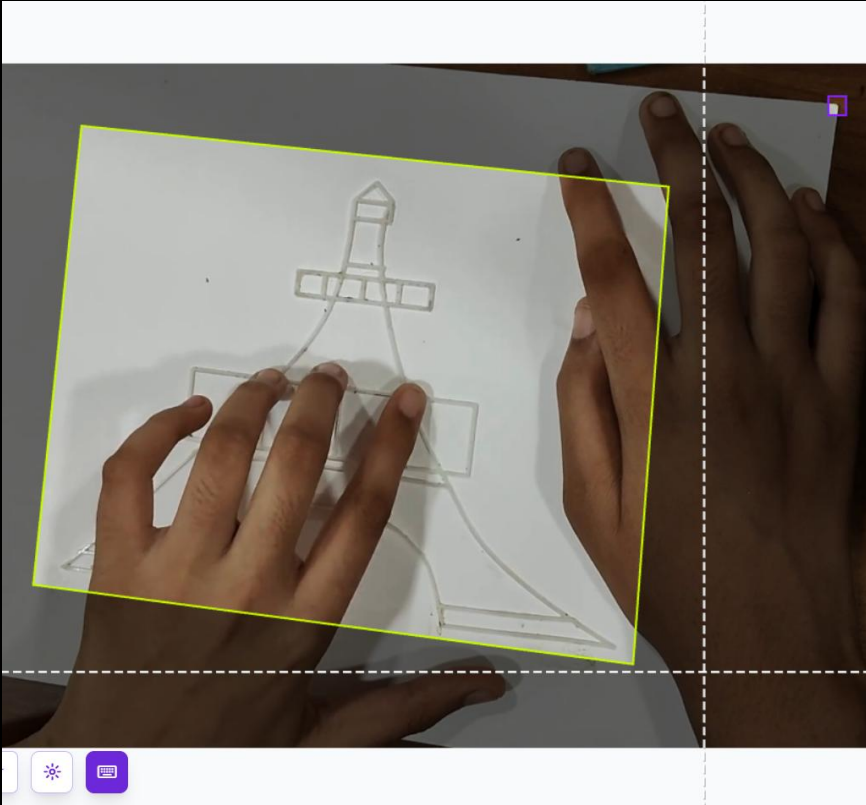
Approximately 70–80 representative frames were extracted from participant videos for each tactile graphic, ensuring that different hand positions, finger orientations, and exploration stages were adequately represented.

These images were uploaded to **Roboflow**, where each finger was manually annotated using bounding boxes.

Roboflow - It is a computer vision platform used for image annotation, dataset management, augmentation, and YOLO-ready dataset generation.



RoboFlow : Data Annotating



Roboflow was further utilized to generate training, validation, and testing datasets while simultaneously applying multiple augmentation techniques such as rotation, flipping, scaling, brightness adjustments, and other geometric transformations.

These **augmentations** improved the robustness of the dataset and enabled the model to generalize across varying hand poses and exploration conditions.

Dataset Preparation with Roboflow

1

Frame Extraction

≈ 70–80 representative frames extracted per tactile graphic, covering diverse finger positions and exploration stages.

2

Manual Annotation

Frames uploaded to Roboflow; finger positions annotated with bounding boxes and verified for quality.

3

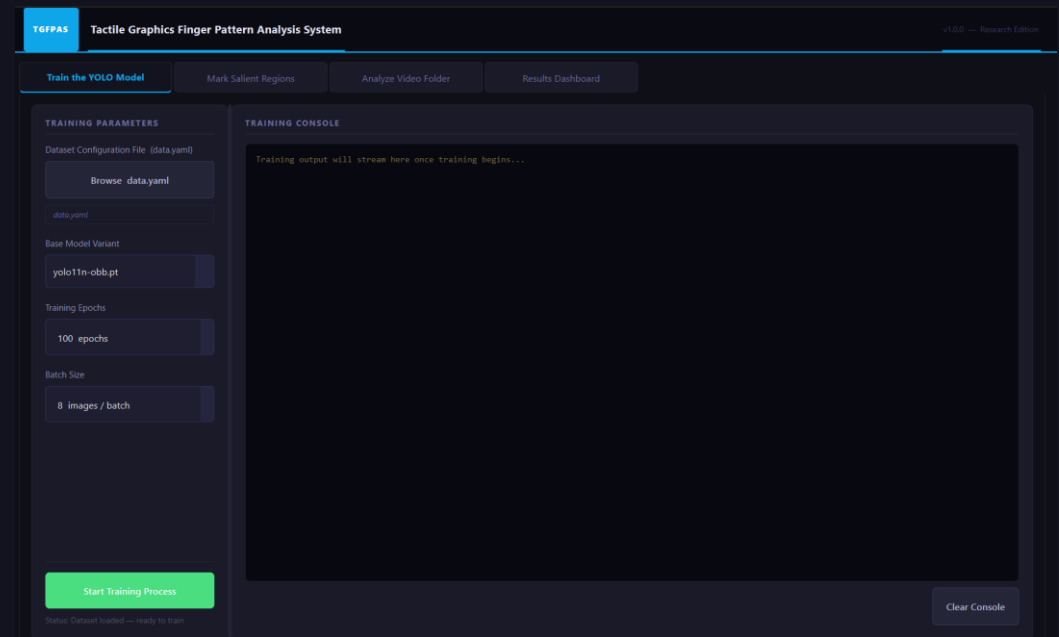
Data Augmentation

Rotation, flipping, scaling, brightness, noise & geometric transforms — generalizing across hand orientations and styles.

4

Dataset Generation

Roboflow auto-exports Train / Validation / Test splits plus the ready-to-use data.yaml configuration file.



Training tab: the user only browses to data.yaml — model variant (yolo11n-obb.pt), epochs (100) and batch size (8) are configurable in the panel.

Why fine-tune instead of a pretrained model? Fingers over raised-line sheets differ heavily from generic detection data — fine-tuning is what makes detection reliable in this domain.

One-Click Training Inside the Application

Train the YOLO Model

Mark Salient Regions

Analyze Video Folder

Results Dashboard

TRAINING PARAMETERS

Dataset Configuration File (data.yaml)

Browse data.yaml

data.yaml

Base Model Variant

yolo11n-obb.pt

Training Epochs

100 epochs

Batch Size

8 images / batch

TRAINING CONSOLE

=====

TRAINING PROCESS STARTED

=====

Dataset : C:/Users/Divyank/Downloads/flower_obb/data.yaml

Model : yolo11n-obb.pt

Epochs : 100

Batch : 8

Dataset root: C:/Users/Divyank/Downloads/flower_obb

Using device: cpu

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Live training console — Ultralytics output streams in real time (dataset scan: 108 train / 10 val images).

BASE MODEL

yolo11n-obb.pt (oriented bounding boxes)

CONFIG

100 epochs · batch 8 · device auto

OUTPUT

best.pt saved automatically per graphic

EXTRAS

Training statistics + validation metrics

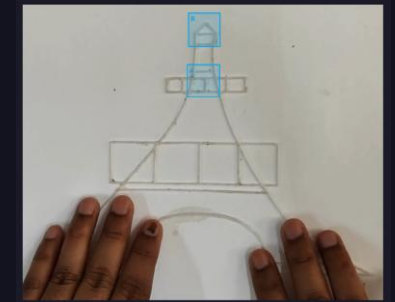
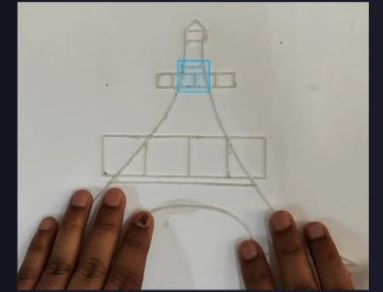
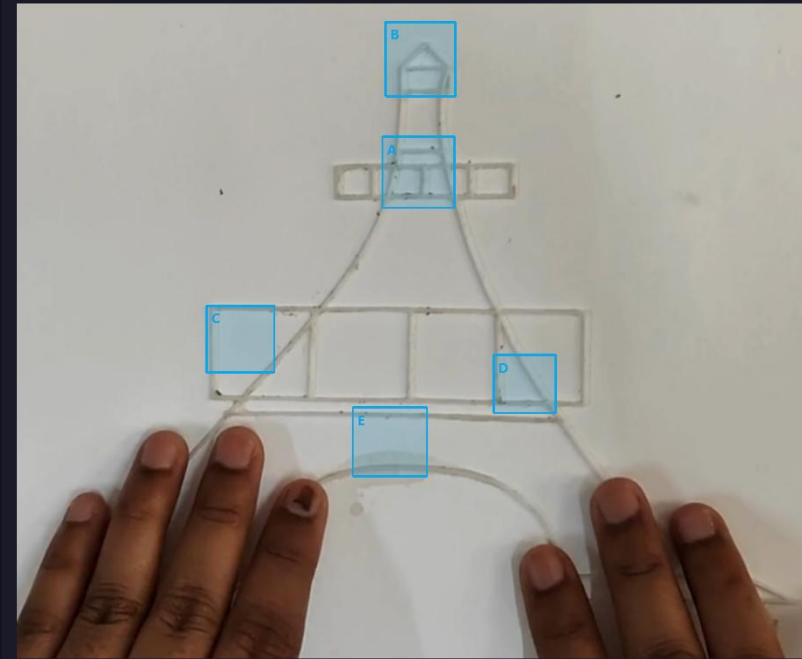
Result: every tactile graphic ends Step 1 with its own dedicated, domain-adapted finger detector — no shared model compromises, no manual command-line work.

Marking Semantically Important Regions

- A clean, warped reference image of the tactile graphic is loaded into the annotation canvas.
- The researcher draws regions of interest directly on the diagram — each becomes a labeled semantic region (A, B, C, D, E...).
- Regions are ranked by expected importance: $A > B > C > D > E$.
- All region IDs, labels and coordinates are saved to regions.json — the single reference map used by every later analysis.

3-step guided workflow in the UI:

Load Reference Image → Draw Regions of Interest → Save Region Map (JSON)



Incremental annotation: regions are added one by one (right) until the full A–E map is complete (left).

Eiffel Tower tactile graphic

Configuring the Analysis Pipeline

TGFPAS

Tactile Graphics Finger Pattern Analysis System

v1.0.0 — Research Edition

Train the YOLO Model

Mark Salient Regions

Analyze Video Folder

Results Dashboard

INPUT CONFIGURATION

Video Folder
INCORRECT

YOLO Model
best.pt

Region Map
regions.json

Tactile Graphic
Screenshot 2025-06-11
004913.png

Run Full Analysis Pipeline

PIPELINE STATUS

All inputs configured — ready to run.

0%

ANALYSIS LOG

Analysis log output will appear here once the pipeline begins...

Clear Log

All four inputs configured — the pipeline is armed and ready to run.

Video Folder

CORRECT or INCORRECT cohort — processed independently

YOLO Model

best.pt from Step 1

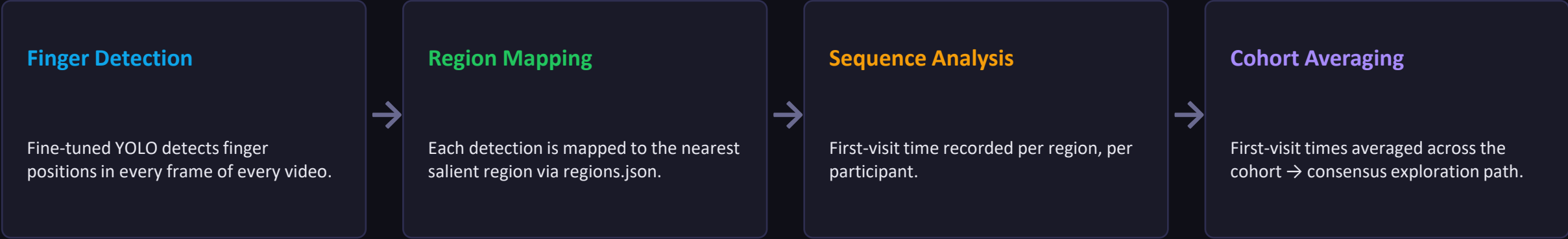
Region Map

regions.json from Step 2

Tactile Graphic

Optional photo used as the spatial-graph background

The Core Behavioral Analysis Engine



HOW FIRST-VISIT TIME WORKS

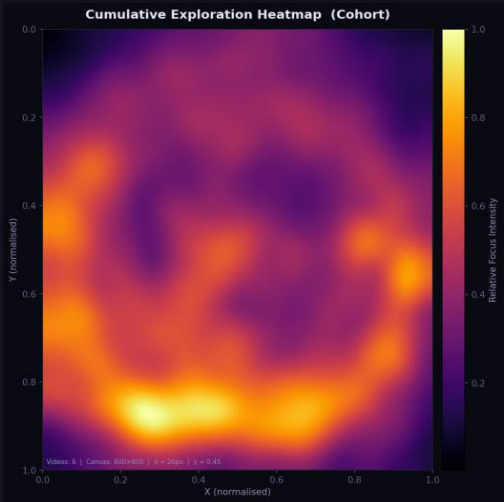
Video 1: A → 2 s B → 5 s C → 9 s D → 14 s
Video 2: A → 3 s C → 6 s B → 8 s D → 11 s
...averaged across the cohort:
A : 1.8 s B : 4.1 s C : 6.7 s D : 12.0 s

Consensus exploration path: A → B → C → D

First-visit time captures the natural strategy, unbiased by repeated touches.

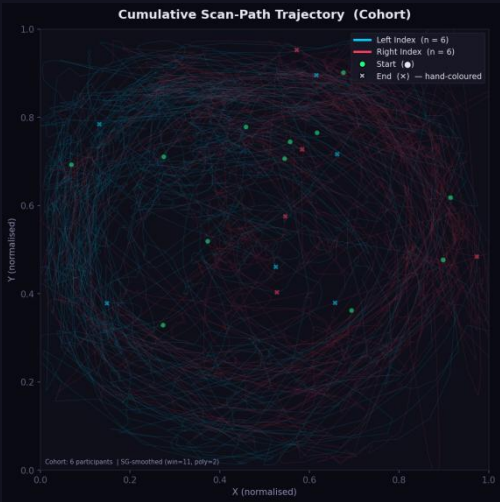
Live pipeline: 20 videos detected, per-video progress, full analysis log.

Automatically Generated Research Analytics – Tire results



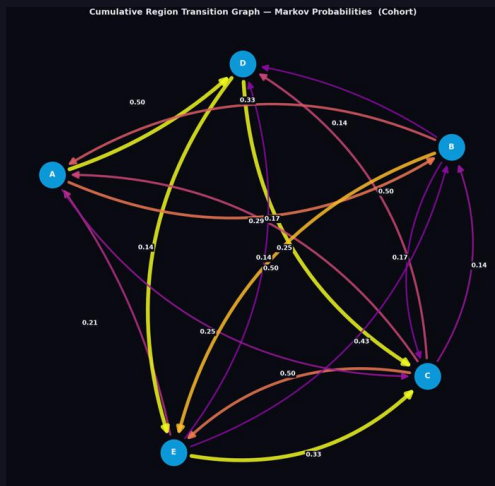
Heatmap

Where attention concentrates on the graphic



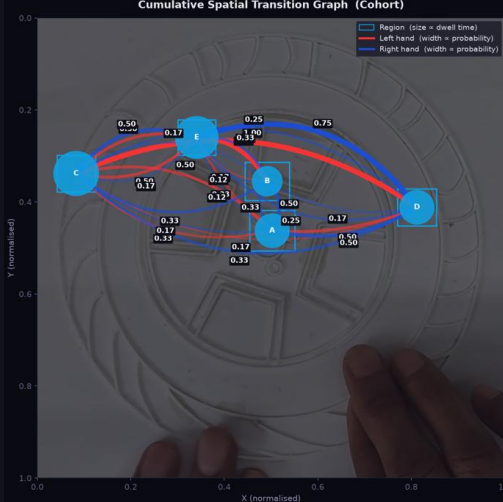
Trajectory Map

Complete finger movement scan-paths



Transition Graph

Probability of moving between regions



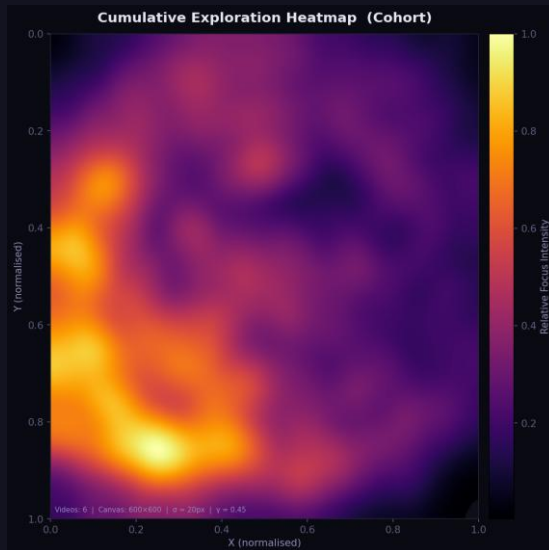
Probabilistic Path

Probability of moving between regions, super imposed on tactile graphics

Plus: spatial transition graphs on the tactile photo · region visit statistics (count, dwell, frequency) · consensus exploration paths · cohort_summary.json with every metric

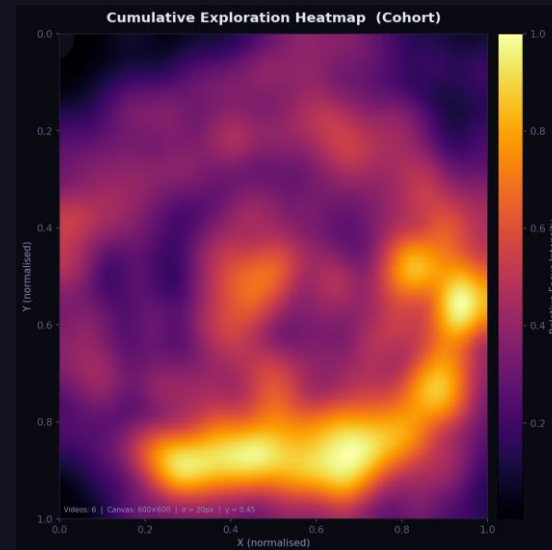
Left Hand · Right Hand · Both Hands — Analyzed Independently

LEFT HAND



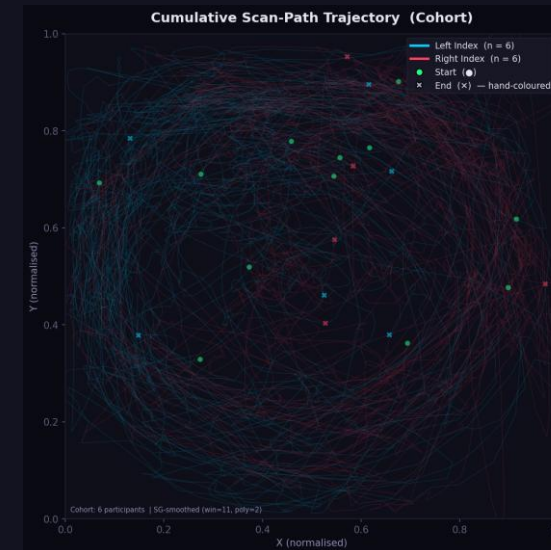
Independent cohort heatmap for this hand

RIGHT HAND



Independent cohort heatmap for this hand

BOTH HANDS (combined)



Left (blue) & right (red) trajectories overlaid in distinct colors

×2
cohorts

Correct &
Incorrect

Why separate hands? Reveals hand dominance, division of labor, and bimanual coordination — e.g., one hand anchoring a landmark while the other scans. Every plot is produced 6× per run: {left, right, both} × {correct, incorrect}.

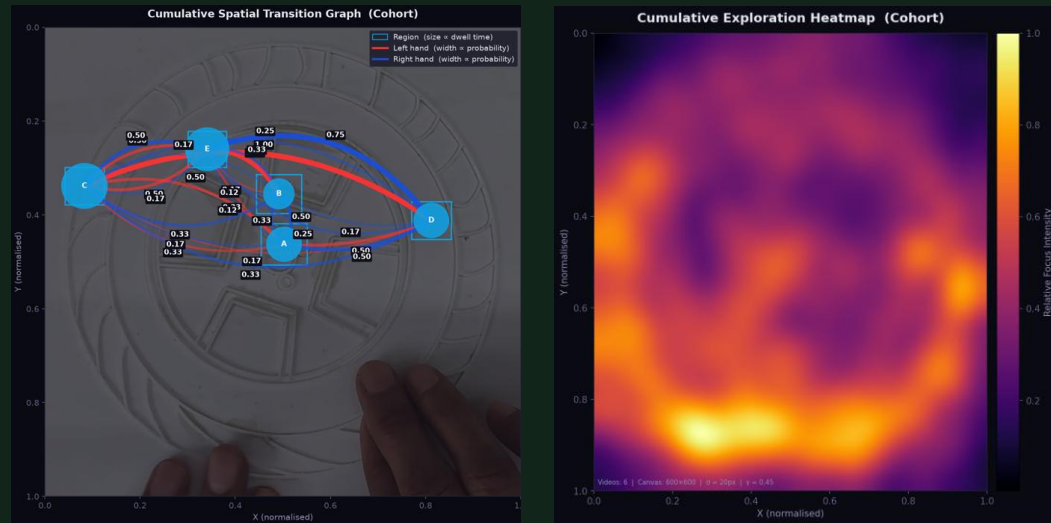
Correct vs Incorrect — The Numbers Tell the Story

Metric	CORRECT (n = 6)	INCORRECT (n = 20)	
Avg exploration time / video	39.8 s	100.4 s	2.5× faster when successful
Total region transitions	8	281	35× fewer back-and-forth switches
Consensus exploration path	B → D → A → C → E	A → B → E → D → C	Fundamentally different strategies
Total exploration time (cohort)	238.9 s	2008.5 s	Efficient, decisive exploration

Metrics computed automatically by TGFPAAS (cohort_summary.json, both-hands analysis).

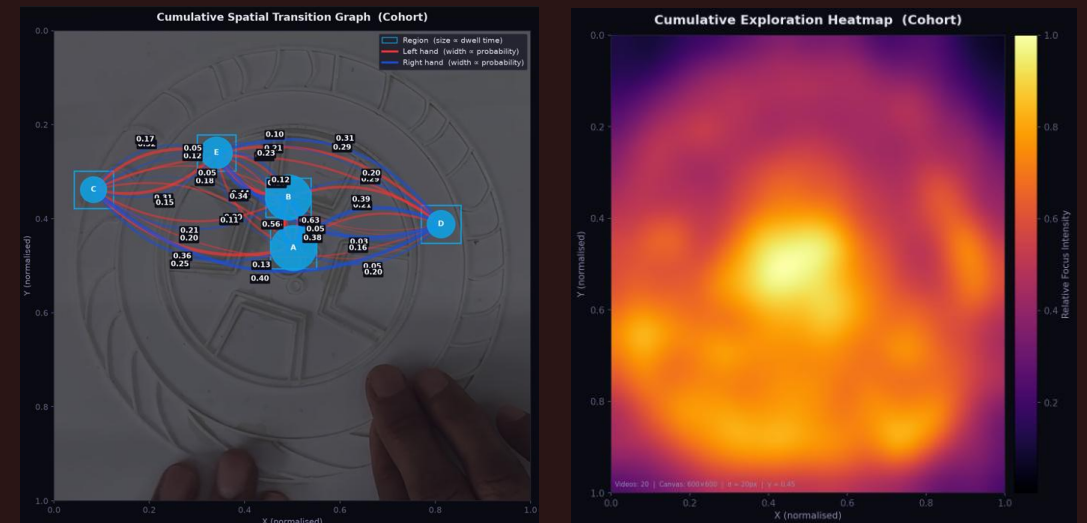
The Behavioral Signature of Success

CORRECT — clean, targeted exploration



Few transitions, focused dwell on discriminative regions.

INCORRECT — dense, wandering exploration



281 transitions criss-crossing the sheet — repeated revisits without convergence.

Key research finding: successful participants explore less but more strategically — efficiency, not effort, predicts correct identification.

Results Dashboard — Explore Everything Interactively

TGFPAS

Tactile Graphics Finger Pattern Analysis System

v1.0.0 — Research Edition

Train the YOLO Model

Mark Salient Regions

Analyze Video Folder

Results Dashboard

Load Output Folder

Export PDF Report

Download CSV Data

SESSION

both_hands

left_hand

right_hand

VISUALIZATION

Plot Type

cumulative_trajectory.png

Load Session Summary

PLOT VIEWER

Cumulative Scan-Path Trajectory (Cohort)

Zoom Annotate

SESSION ANALYTICS

Videos Processed

Total Exploration (s)

Avg Per Video (s)

Total Region Transitions

REGION METRICS

Region	Avg First Visit (s)	Total Dwell Frames

RAW SESSION SUMMARY

Session browser (both/left/right) · plot viewer with zoom & annotate · session analytics · region metrics table.

Zero recomputation

Loads saved Step-3 outputs directly from the output folder.

Session analytics

Videos processed, total & average exploration time, region transitions.

Region metrics

Avg first-visit time and total dwell frames per region.

One-click export

PDF report + CSV data download for further statistical analysis.

Technology Stack



Python

Core language for the entire pipeline



PySide6

Multi-tab desktop GUI with threaded workers



Ultralytics YOLO11

OBB finger detection & in-app fine-tuning



Roboflow

Annotation, augmentation & dataset export



OpenCV

Frame extraction & video processing



PyTorch

Model training backend



NumPy · Pandas

Metric computation & data wrangling



Matplotlib

Heatmaps, trajectories & graph rendering



JSON

Region maps & cohort summaries

Graph-based modeling for transitions & trajectories · modular tab architecture (training / region / analysis / results) mirrored in code.

What Makes This a Research Framework, Not Just an App

- Automated, per-graphic YOLO fine-tuning — each tactile graphic gets its own domain-adapted detector.
- JSON-based semantic region mapping with ranked importance ($A > B > C > D > E$).
- Frame-wise finger detection with region assignment across entire video cohorts.
- First-visit-time temporal analysis → unbiased consensus exploration strategies.
- Independent Correct vs Incorrect cohort processing for behavioral comparison.
- Separate Left / Right / Combined hand analysis with color-coded bimanual visualization.
- Full analytics suite: heatmaps, trajectories, transition matrices, probabilistic movement graphs, direction analysis, region statistics.
- Interactive dashboard with PDF & CSV export — no recomputation needed.
- Scalable by design: add a new graphic = new best.pt + new regions.json. Zero code changes.
- Relevant to HCI, accessibility, computer vision & assistive-technology research.

CONTRIBUTIONS

ARYAN (Application development)

- Reviewed the existing application, identified UI, workflow, and functionality gaps, and used those insights to refine the four-stage pipeline; fine-tuning, region annotation, analysis, results.
- Restructured the analysis module from individual to cumulative, whole cohort results and unified the interface with a global stylesheet.
- Built the transition-graph overlay including superimposed representation and final consensus sequence.
- Explored a frame-capture-threshold optimization to cut processing time of the application during its testing phase.

AVDHESH (Annotation & training data)

- Studied the related research paper to inform the annotation methodology.
- Annotated all 8 tactile graphics to build the training dataset; which is the most important step before YOLO finetuning.
- Prepared the data pipeline, frame extraction, bounding-box annotation, augmentation, dataset splits.
- Fixed a normalization mismatch between the annotation and results stages.
- Proposed and helped develop a new methodology for marking salient regions, based on frequency-visit ordering of transition hubs

DIVYANK (Application & annotation support)

- Worked across both the application (analysis and results stages) and annotation (annotated 4 datasets)/training-data preparation the only team member spanning both halves of the pipeline.
- Co-designed the Left/Right/Both-hand analysis, including the recommendation to separate hand paths by color in the combined view.
- Carried out annotation work across multiple tactile graphics.
- Led the majority of testing and validation across the team, and help out in figuring out the path calculation for both correct and incorrect videos.

Model training & testing - All three members, Trained and fine-tuned all 12 YOLO detection models together across sessions in the Gaming and Animation Laboratory. Jointly tested all trained models and generated the testing results. Ran the full application testing, validation, and usability study together. Documented individual contributions and the overall project workflow

From Raw Videos to Exploration Insight — Fully Automated

4

independent modules covering the full research workflow

6×

analytics per run — {left, right, both} × {correct, incorrect}

2.5×

faster exploration observed in successful participants

0

code changes needed to extend to a new tactile graphic

Future scope

More tactile graphics & larger cohorts · statistical significance testing between cohorts · real-time exploration feedback for training BVI users · fingertip-level (multi-finger) tracking · automatic salient-region suggestion from touch density.

Thank You!

Tactile Graphics Finger Pattern Analysis System · Questions welcome